

TMA4268 Statistical Learning V2026

Mock Exam 5 (estimated final exam)

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Based on the Apr 28 exam-review lecture, the 2023–2025 finals, and the prof’s stated scope rule

Mock for: May 18, 2026 (real exam date)

Instructions.

- **Duration:** 4 hours. **Open book.** Permitted aids: ISLP (2nd ed.), one handwritten A5 sheet of notes, calculator.
- **No code required.** Write answers as math, plain English, or pseudocode. Do *not* memorize R/Python package names.
- **Show your work** — partial credit is generously available, including when calculator slips spoil a numeric answer but the setup is correct.
- **No negative scoring.** Always answer, even when unsure.
- **If a question seems broken or ambiguous**, state the assumption you are making in one short sentence and proceed.
- **Total: 100 points = 100 %.** Per-problem weights given in parentheses.

Grade boundaries (NTNU *prosentvurderingsmetoden*, advisory): A: 89–100 % B: 77–88 % C: 65–76 % D: 53–64 % E: 41–52 % F: 0–40 %.

Problem 1 (10 %) — Fill-in-the-blank concepts

Read the passage and pick the best word or short phrase for each blank from the choices in parentheses. Each correct fill is worth 1 %.

In multiple linear regression, when two or more predictors are highly correlated with each other, the matrix $\mathbf{X}^\top \mathbf{X}$ becomes near-singular and the variances of the estimated coefficients $\hat{\boldsymbol{\beta}}$ inflate. This phenomenon is known as _____ (1) (*collinearity* / *heteroscedasticity* / *leverage* / *influence*). Two classical remedies discussed in this course are ridge regression, which adds a penalty proportional to the squared L2 norm of the coefficient vector and is therefore also called a _____ (2) (*forward-selection* / *shrinkage* / *decorrelation* / *bagging*) method, and principal components regression, which first replaces the correlated predictors by their orthogonal _____ (3) (*principal components* / *residuals* / *Tukey directions* / *leverages*) and then regresses on a subset of those.

In neural networks, fitting a feedforward model with many more parameters than training observations is only possible because we always combine training with one or more forms of regularization. Common explicit choices include L1 / L2 weight decay; randomly zeroing a fraction of node outputs at each training iteration, called _____ (4) (*pruning* / *pooling* / *dropout*

/ *thinning*); halting training when validation error stops improving, called _____ (5) (*warm starting* / *early stopping* / *tolerance gating* / *patience tuning*); and softening one-hot targets so the network does not become over-confident, called _____ (6) (*label smoothing* / *soft margins* / *class reweighting* / *batch normalization*). In addition, training the network with stochastic gradient descent on _____ (7) (*full passes* / *leave-one-out folds* / *out-of-bag samples* / *mini-batches*) provides an *implicit* regularization effect for free, which is one of the reasons over-parameterized networks generalize.

For estimating the test error of a chosen procedure honestly while also using cross-validation to *select* a hyperparameter, the recommended approach is to run two layers of cross-validation, with outer folds for assessment and inner folds for selection. This procedure is known as _____ (8) (*nested cross-validation* / *stratified CV* / *leave-one-out CV* / *bootstrap aggregation*).

In ensemble methods for classification, _____ (9) (*bagging* / *AdaBoost* / *random forests* / *LDA*) builds a sequence of weak classifiers, each fit on training data that have been *re-weighted* so that observations the previous classifiers have misclassified receive higher weight. A modern unifying view interprets this and related methods as fitting a tree at each step to the *negative gradient* of a chosen differentiable loss, giving the general framework called _____ (10) (*gradient boosting* / *coordinate descent* / *Newton stepping* / *posterior averaging*).

Problem 2 (28 %) — Multiple choice, true/false, and short numeric

For each subproblem, write *True* or *False* for each statement (or the requested numeric answer). You may add a one-sentence justification, but only if you think it helps; do not write essays.

a) Bias–variance decomposition and double descent (3 %)

Mark each statement true or false.

- (i) In the decomposition $\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \text{Bias}^2 + \text{Var} + \sigma^2$, the irreducible term σ^2 can be made smaller by collecting more training data.
- (ii) In the over-parameterized regime ($p \gg n$), the bias–variance decomposition no longer holds as an algebraic identity, which is why test MSE can exhibit a second descent past the interpolation peak.
- (iii) For a fixed problem and a fixed model class, switching from no regularization to a well-tuned regularizer typically lowers variance *more* than it raises bias, and therefore lowers expected test error.

b) Cross-validation and its pitfalls (3 %)

Mark each statement true or false.

- (i) Leave-one-out CV (LOOCV) has *lower* bias than 10-fold CV as an estimator of test error, because each LOOCV fit uses $n - 1$ observations.
- (ii) In nested cross-validation, the *inner* folds are used to select a hyperparameter (or model), and the *outer* folds are used to give an honest estimate of the chosen procedure’s test error.
- (iii) For temporally autocorrelated data, randomly assigning observations to k folds gives an *honest* estimate of test error provided k is large enough (say $k \geq 10$).

c) Bootstrap, standard errors and CIs (3 %)

- (i) (1 %) For a training sample of size $n = 10$, what is the probability (rounded to three decimals) that a given original observation is included in a particular bootstrap sample?
- (ii) (1 %) You wish to obtain a 95% confidence interval for a scalar statistic $\hat{\theta}$ using $B = 1000$ bootstrap resamples. Specify the *percentile* confidence interval as two specific quantiles of the bootstrap distribution $\{\hat{\theta}_1^*, \dots, \hat{\theta}_B^*\}$.
- (iii) (1 %) Mark each of the following true or false. (A) The bootstrap standard error of $\hat{\theta}$ is the sample standard deviation across the bootstrap replicates $\hat{\theta}_b^*$. (B) The bootstrap can correct for the bias of $\hat{\theta}$ as an estimator of θ . (C) Bootstrap resamples should be drawn *without* replacement to avoid duplicating observations.

d) Neural network regularization (3 %)

Mark each statement true or false.

- (i) In a feedforward network trained with dropout, the dropout layer is active during both training and test-time prediction.
- (ii) Label smoothing replaces hard one-hot targets like $(0, 0, 1, 0)$ with softened targets like $(\varepsilon/3, \varepsilon/3, 1 - \varepsilon, \varepsilon/3)$ for a small $\varepsilon > 0$; this tends to make the model less over-confident on the training set and improves generalization in classification tasks.
- (iii) Adding L2 weight decay decreases the network’s training error.

e) Mini-batch SGD and backpropagation (3 %)

Mark each statement true or false.

- (i) Backpropagation is the algorithm that *performs* the parameter update at each iteration of mini-batch SGD, by stepping each weight in the direction that most decreases the per-batch loss.
- (ii) Mini-batch stochastic gradient descent yields, on each update, an *unbiased* estimator of the full-data gradient.
- (iii) In typical practice, batch sizes for mini-batch SGD are chosen as powers of two (e.g. 32, 128, 256) primarily for hardware-efficiency reasons rather than statistical reasons.

f) Boosting: AdaBoost, gradient boosting, XGBoost (4 %)

- (i) (1 %) In the AdaBoost algorithm, at iteration m the weak classifier $G_m(x)$ has weighted misclassification rate $\text{err}_m = 0.20$ on the current sample weights w_i . Using $\alpha_m = \log \frac{1-\text{err}_m}{\text{err}_m}$, compute the resulting classifier weight α_m (two decimals).
- (ii) (1 %) For a gradient boosting model with squared-error loss $L(y, f) = \frac{1}{2}(y - f)^2$, give the negative gradient $-\partial L / \partial f$ evaluated at the current ensemble's prediction $\hat{f}^{(m-1)}(x_i)$, and identify what familiar quantity it equals.
- (iii) (2 %) Mark each statement true or false. (A) In gradient boosting, the number of trees M is a real tuning parameter; using M much larger than necessary will typically over-fit. (B) For a random forest, in contrast, the number of trees B is *not* a serious tuning parameter and one can simply “use enough.” (C) Decreasing the learning rate ν in gradient boosting usually *decreases* the number of trees M needed to reach a good fit. (D) XGBoost augments vanilla gradient boosting with second-order (Newton-like) gradient information and with L1 / L2 penalties on the leaf weights.

g) Logistic regression: odds, log-odds, interaction (3 %)

- (i) (1 %) A logistic regression of **chd** (1 = developed coronary heart disease, 0 = did not) on **ldl** (LDL cholesterol, mmol/L) reports $\hat{\beta}_{\text{ldl}} = 0.30$. By what factor (two decimals) do the odds of CHD change for a 2 mmol/L increase in LDL, holding other predictors fixed?
- (ii) (1 %) A model includes the predictors **age** and **smoker** and an interaction **age:smoker**. The fitted coefficients are $\hat{\beta}_{\text{age}} = 0.04$ and $\hat{\beta}_{\text{age:smoker}} = 0.03$. By what factor (two decimals) do the odds of the event change for a one-year increase in **age** for a *smoker*?
- (iii) (1 %) True or false: “In a binary logistic regression with a one-unit increase in x_j , the *probability* of $Y = 1$ changes by approximately $\hat{\beta}_j$.”

h) PCA: explained variance and loadings (3 %)

You perform PCA on $p = 6$ *standardized* variables and obtain eigenvalues

$$\lambda_1 = 2.7, \lambda_2 = 1.5, \lambda_3 = 0.9, \lambda_4 = 0.5, \lambda_5 = 0.3, \lambda_6 = 0.1.$$

- (i) (1 %) What is the total variance in the standardized data, and what proportion of this total variance is explained by the first two principal components combined (two decimals)?
- (ii) (1 %) How many principal components must be retained to capture at least 90% of the total variance? Show your cumulative sum.
- (iii) (1 %) True or false: “Because the variables were standardized before PCA, the loadings ϕ_{jk} for the k -th principal component depend only on $\mathbf{X}$ and not on the response y .”

i) Discriminant analysis and splines (3 %)

- (i) (1 %) True or false: “Under the LDA assumption that the class-conditional densities share a common covariance matrix Σ , the resulting Bayes-optimal decision boundary between two classes is linear in $\mathbf{x}$.”
- (ii) (1 %) True or false: “The QDA decision boundary is *quadratic* in $\mathbf{x}$ because the class-specific covariances Σ_k make the term $\mathbf{x}^\top \Sigma_k^{-1} \mathbf{x}$ in the discriminant fail to cancel between classes.”
- (iii) (1 %) A cubic regression spline (using the truncated-power basis, with a global intercept) on a single predictor x has K interior knots. How many parameters does this spline term consume, *including* the global intercept of the model?

Problem 3 (16 %) — Theory and pseudocode

a) The mathy one — bias–variance decomposition (7 %)

Let

$$y_0 = f(x_0) + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2, \quad \varepsilon \perp x_0,$$

and let $\hat{f}$ be a model fit on a random training set $\mathcal{D}$, independent of ε . All expectations and variances below are jointly over the random training set $\mathcal{D}$ and the noise ε at the test point.

- (i) (2 %) Show that

$$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] + \sigma^2.$$

Be explicit about why the cross term vanishes.

- (ii) (3 %) Starting from the right-hand side of (i) and the trick of adding and subtracting $\mathbb{E}[\hat{f}(x_0)]$, derive

$$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = (f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2 + \text{Var}(\hat{f}(x_0)) + \sigma^2.$$

Identify each term with its standard name (*Bias*², *Variance*, *Irreducible error*) and explain in one sentence which randomness each term captures.

- (iii) (1 %) A lasso regression is fit with λ chosen by CV. Compared with ordinary least squares on the same data, the lasso typically has *higher* bias and *lower* variance. In one short sentence each, explain (a) why the bias is generally higher under lasso, and (b) why the variance is generally lower.
- (iv) (1 %) A student writes: “Bias–variance is just a *trade-off*; you cannot improve both at the same time.” Briefly explain why this assertion is too strong in light of regularization and the over-parameterized / double-descent regime. (One short sentence.)

b) Cross-validation: pseudocode and nested CV (4 %)

- (i) (2 %) Write *pseudocode* (math, plain English, or programming-language-agnostic notation) for k -fold cross-validation of a regression model $\mathcal{M}$ at a fixed hyperparameter setting θ , returning the cross-validation MSE estimate $\widehat{\text{CV}}_k(\theta)$. Make explicit (a) how the data is partitioned, (b) what is held out and what is fit at each iteration, and (c) how the per-fold errors are aggregated.
- (ii) (1 %) Using the pseudocode in (i) as a primitive, write a short description of how to use it to *choose* a hyperparameter θ from a finite grid $\theta \in \{\theta_1, \dots, \theta_G\}$, and how to use the chosen θ to produce a final fitted model.
- (iii) (1 %) A colleague proposes the following pipeline: “Step 1: compute the marginal correlation of each of $p = 5000$ predictors with the response y on the full training set; keep the top 25. Step 2: run 10-fold CV on a logistic regression fit using just those 25. The CV misclassification rate is my honest estimate of test error.” Briefly state why this estimate is biased *downward*, and describe in one or two sentences the corrected procedure.

c) Bootstrap by hand (3 %)

- (i) (1 %) Show that, for a sample of size n , the probability that a specific observation i is *not* included in a given bootstrap sample is $(1 - 1/n)^n$, and that this probability tends to $1/e \approx 0.368$ as $n \rightarrow \infty$. (One line of reasoning is sufficient.)
- (ii) (1 %) You wish to estimate the standard error of the sample median $\hat{\theta} = \widehat{\text{med}}(X_1, \dots, X_n)$, for which no clean closed-form expression is available. Describe in 3–4 short sentences (or short pseudocode lines) the bootstrap procedure that estimates $\widehat{\text{SE}}_{\text{boot}}(\hat{\theta})$ using B resamples. Be explicit about whether resampling is with or without replacement.
- (iii) (1 %) A logistic regression of **default** on **balance** produces $\hat{\beta}_{\text{balance}} = 0.005$ with closed-form $\widehat{\text{SE}}(\hat{\beta}) = 0.00020$ from the inverse Fisher information. A bootstrap with $B = 5000$ resamples gives $\widehat{\text{SE}}_{\text{boot}}(\hat{\beta}) = 0.00050$. Give *one* substantive reason (one sentence) why these two SEs can disagree.

d) Hierarchical clustering by hand (2 %)

Four observations have the following Euclidean dissimilarity matrix:

$$D = \begin{pmatrix} 0 & 2 & 8 & 7 \\ 2 & 0 & 9 & 6 \\ 8 & 9 & 0 & 3 \\ 7 & 6 & 3 & 0 \end{pmatrix}.$$

- (i) (1 %) Run agglomerative hierarchical clustering with *complete* linkage. List the two earliest fusions (which observations / clusters fuse, and at what height).
- (ii) (1 %) At what height does the *final* fusion (all four observations joined into one cluster) occur under complete linkage? Show the inter-cluster distance you used.

Problem 4 (22 %) — Data analysis: energy use of homes

An energy utility collects $n = 800$ single-family houses' annual heating cost **cost** (in 1000 EUR). The available predictors are:

- **area** — floor area (m^2 , continuous);
- **age** — age of the house (years, continuous);
- **insulation** — categorical, 3 levels: *poor* (reference), *standard*, *premium*;
- **heatpump** — binary 0/1: 1 if the house uses a heat pump, 0 otherwise;
- **rooms** — number of rooms (count).

The data are split 600/200 into a training and a test set. All continuous predictors are standardized to mean 0, standard deviation 1 *before* fitting any of the models below; the standardized variables are denoted $\widetilde{\text{area}}$, $\widetilde{\text{age}}$, $\widetilde{\text{rooms}}$ (so that one unit in the model corresponds to one standard deviation in the raw variable).

a) Linear regression with a polynomial and an interaction (7 %)

The course staff first fit, on the training set, the model

$$\text{cost} \sim \widetilde{\text{area}} + \widetilde{\text{age}} + \widetilde{\text{age}}^2 + \text{insulation} + \text{heatpump} + \widetilde{\text{area}}:\text{heatpump} + \widetilde{\text{rooms}}.$$

The fitted output is:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	2.80	0.10	28.00	< 0.001
$\widetilde{\text{area}}$	0.80	0.09	8.89	< 0.001
$\widetilde{\text{age}}$	-0.05	0.10	-0.50	0.617
$\widetilde{\text{age}}^2$	0.30	0.06	5.00	< 0.001
insulation_standard	-0.40	0.10	-4.00	< 0.001
insulation_premium	-0.80	0.12	-6.67	< 0.001
heatpump	-0.60	0.15	-4.00	< 0.001
$\widetilde{\text{area}}:\text{heatpump}$	-0.50	0.12	-4.17	< 0.001
$\widetilde{\text{rooms}}$	0.10	0.09	1.11	0.268

Multiple $R^2 = 0.71$, Adjusted $R^2 = 0.706$. Residual standard error: 0.45 on 591 d.f.

- (1 %) How many parameters (including the intercept) does this model estimate? Verify the count against the printed residual degrees of freedom.
- (2 %) For a house *without* a heat pump, by how much does an increase of one standard deviation in $\widetilde{\text{area}}$ change the predicted **cost** (in 1000 EUR), holding all other predictors fixed? Repeat the calculation for a house *with* a heat pump, and briefly comment on the sign of the difference.
- (2 %) Consider two houses, both at $\widetilde{\text{area}} = 0$, $\widetilde{\text{rooms}} = 0$, **insulation** = *poor*, **heatpump** = 0. House *A* has $\widetilde{\text{age}} = -1$ (one SD younger than average); house *B* has $\widetilde{\text{age}} = +2$ (two SDs older than average). Compute the predicted **cost** of each (two numerics, 1000 EUR). Briefly comment on what the $\widetilde{\text{age}}^2$ term implies about the marginal effect of age at older versus younger houses.

- (iv) (1 %) A classmate writes: “ $\widetilde{\text{age}}$ has a p -value of 0.617, so we should drop it from the model.” In one or two sentences, explain why this is wrong given the rest of the fitted model. (Hint: marginal effect of $\widetilde{\text{age}}$ depends on both $\hat{\beta}_{\widetilde{\text{age}}}$ and $\hat{\beta}_{\widetilde{\text{age}}^2}$, and this course follows the hierarchical-principle convention.)
- (v) (1 %) The main-effect coefficient on **heatpump** is $\hat{\beta}_{\text{heatpump}} = -0.60$. A classmate concludes: “On average, houses with a heat pump cost €600 per year less to heat than houses without.” Briefly explain why this is misleading *given the interaction in the model*, and write down the actual change in predicted **cost** (in 1000 EUR) of installing a heat pump for a house at standardized area $\widetilde{\text{area}} = +1$.

b) Collinearity diagnosis (4 %)

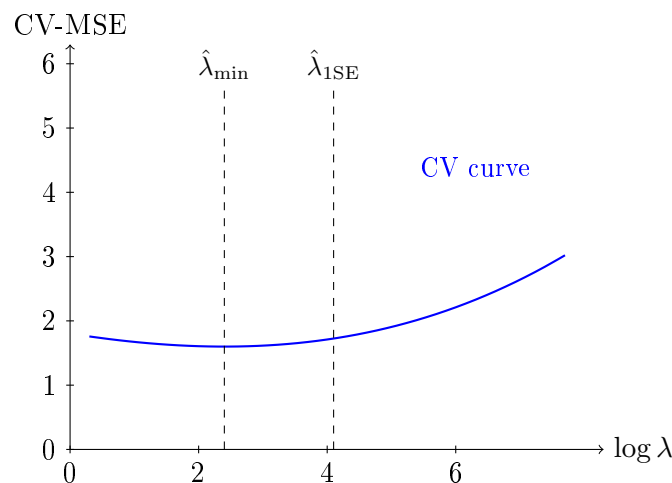
In a separate, more realistic version of the same dataset, the predictors $\widetilde{\text{area}}$ and $\widetilde{\text{rooms}}$ are found to have empirical correlation $r = 0.92$. The course staff fit two models, both *without* the interaction and quadratic terms, on the same training set:

Model	$\hat{\beta}_{\widetilde{\text{area}}}$	SE	$\hat{\beta}_{\widetilde{\text{rooms}}}$	SE	Adjusted R^2
$\widetilde{\text{area}}$ only	0.90	0.06	—	—	0.43
$\widetilde{\text{area}} + \widetilde{\text{rooms}}$	0.55	0.15	0.42	0.16	0.44

- (i) (1 %) Briefly identify the *two* numerical symptoms in the second row of the table above that are characteristic of collinearity.
- (ii) (1 %) Explain in one short sentence the source of this behavior in terms of $(\mathbf{X}^\top \mathbf{X})^{-1}$ and the fact that $\widetilde{\text{area}}$ and $\widetilde{\text{rooms}}$ are nearly the same column of the design matrix.
- (iii) (1 %) Briefly comment on whether *predictions* of **cost** from the second model are also expected to be unstable, given that the joint contribution $\hat{\beta}_{\widetilde{\text{area}}} \cdot \widetilde{\text{area}} + \hat{\beta}_{\widetilde{\text{rooms}}} \cdot \widetilde{\text{rooms}}$ is well-determined.
- (iv) (1 %) Name two methods discussed in this course that one could apply to address the collinearity *without* simply dropping one of the two predictors, and for each one state in one short sentence the trade-off it makes (in terms of bias, variance, or interpretability).

c) Lasso with 10-fold CV (5 %)

The same $p = 8$ *standardized* predictors from part (a) (including the $\widetilde{\text{age}}^2$ and the $\widetilde{\text{area}}:\text{heatpump}$ interaction) are now fed into a lasso regression. A 10-fold cross-validation gives the CV-MSE profile



The corresponding test MSEs on the held-out 200 observations:

	Test MSE	No. of nonzero coefs (excl. intercept)
OLS (full model from (a))	0.205	8
Lasso at $\hat{\lambda}_{\min}$	0.196	6
Lasso at $\hat{\lambda}_{1SE}$	0.203	4

- (i) (1 %) The `insulation` factor has been entered into the design matrix as two dummies (`insulation_standard` and `insulation_premium`). Briefly explain why the resulting lasso path is *not* invariant to the choice of reference level (i.e. which level is dropped). What practical consequence does this have for interpretation when one of the dummies is shrunk to exactly zero?
- (ii) (1 %) State the *one-standard-error rule* precisely, and give one reason a practitioner might choose $\hat{\lambda}_{1SE}$ over $\hat{\lambda}_{\min}$.
- (iii) (1 %) On this dataset, lasso at $\hat{\lambda}_{\min}$ improves test MSE from 0.205 to 0.196, a small improvement, and drops 2 of 8 coefficients. Interpret in bias–variance terms: what does the small improvement plus a modest reduction in active predictors suggest about the importance of the dropped predictors?
- (iv) (2 %) Suppose the practitioner cares primarily about *interpretability* and would prefer a simpler model. Which of $\hat{\lambda}_{\min}$ and $\hat{\lambda}_{1SE}$ would you recommend, and why? Give a concrete claim about the trade-off the recommendation makes in test MSE terms (refer to the numbers in the table).

d) Gradient boosting interpretation and pseudocode (6 %)

A gradient-boosted regression tree (squared-error loss) is fit on the same training data, achieving test MSE = 0.151, lower than the lasso.

- (i) (2 %) Write *pseudocode* for the squared-error gradient-boosting algorithm, with B trees, learning rate ν and tree depth d as hyperparameters. Make explicit (a) the initialization, (b) the per-iteration step (a fit-residuals view is fine for squared-error loss), and (c) the final returned function. Two short sentences of accompanying explanation are fine if helpful.
- (ii) (1 %) For *squared-error* loss, derive in one or two lines that fitting a tree to the current residuals $r_i^{(m)} = y_i - \hat{f}^{(m-1)}(x_i)$ is equivalent to fitting a tree to the negative gradient of the loss with respect to the function values $\hat{f}^{(m-1)}(x_i)$.
- (iii) (1 %) Briefly describe how you would tune the three boosting hyperparameters (B, d, ν) in practice, and tie each one’s role to a single sentence about bias–variance.
- (iv) (1 %) A junior colleague proposes setting $B = 10,000$ “just to be safe.” Is this a good idea? Why or why not?
- (v) (1 %) The gradient boosting test MSE (0.151) is substantially below the OLS / lasso test MSE (≈ 0.20). Briefly interpret what this gap suggests about the structure of the underlying relationship between predictors and `cost`, and name one diagnostic plot you would compute from the boosted ensemble to recover some *interpretability*.

Problem 5 (24 %) — Data analysis: hospital-readmission classification

A regional hospital records $n = 3000$ patients discharged after a cardiac event and follows each for 90 days. The binary response `readmit` is 1 if the patient is readmitted within 90 days and 0 otherwise. The available predictors are:

- `age` — age in years (continuous);
- `los` — length of initial hospital stay (days, continuous);
- `prev_adm` — number of prior admissions in the past year (count, 0–10);
- `diabetes` — binary 0/1 indicator;
- `insulin` — binary 0/1 indicator (insulin therapy during stay);
- `sex` — binary 0/1 (female = 0 reference, male = 1).

The data are split 70/30 into training (2100) and test (900) sets. Among the test set, 180 patients are truly readmitted and 720 are not.

a) Logistic regression with an interaction (7 %)

A logistic regression model is fit on the training set:

$$\text{logit}(\Pr(\text{readmit} = 1 \mid X)) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{los} + \beta_3 \text{prev_adm} + \beta_4 \text{diabetes} + \beta_5 \text{insulin} + \beta_6 (\text{diabetes} : \text{insulin})$$

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	−4.20	0.45	−9.33	< 0.001
<code>age</code>	0.030	0.005	6.00	< 0.001
<code>los</code>	0.080	0.020	4.00	< 0.001
<code>prev_adm</code>	0.40	0.06	6.67	< 0.001
<code>diabetes</code>	0.10	0.18	0.56	0.578
<code>insulin</code>	0.05	0.20	0.25	0.804
<code>diabetes:insulin</code>	0.90	0.30	3.00	0.003
<code>sex</code>	0.15	0.12	1.25	0.211

- (1 %) State your encoding assumption for the binary predictors `diabetes` and `insulin` explicitly. By what factor (two decimals) do the odds of readmission multiply for each additional prior admission `prev_adm`, holding the other predictors fixed?
- (2 %) For each of the four combinations of $(\text{diabetes}, \text{insulin}) \in \{(0, 0), (1, 0), (0, 1), (1, 1)\}$, write down the joint contribution of the three diabetes / insulin terms to the linear predictor $\hat{\eta}$. By what factor do the odds of readmission change for a diabetic-and-on-insulin patient compared with a reference patient (`diabetes` = 0, `insulin` = 0), holding all other predictors fixed? (Two decimals.)
- (2 %) Consider a 70-year-old male patient with `los` = 6 days, `prev_adm` = 2, `diabetes` = 1, `insulin` = 1. Compute the linear predictor $\hat{\eta}$ and the predicted probability $\hat{p}$ of readmission. Show your work.
- (1 %) A classmate writes: “ $\hat{\beta}_{\text{diabetes}} = 0.10$ with p -value 0.578 and $\hat{\beta}_{\text{insulin}} = 0.05$ with p -value 0.804. Both are insignificant, so we should drop both `diabetes` and `insulin` from the model.” In one or two sentences, explain why this is wrong given that the model also contains an `diabetes:insulin` interaction.
- (1 %) Define, *in plain words appropriate to this readmission setting*, what *sensitivity* means for this classifier. (One sentence.)

b) Confusion matrix and threshold tuning (3 %)

On the held-out test set, the logistic model with default threshold $\hat{p} = 0.5$ produces the confusion matrix

	Predicted: readmit	Predicted: no readmit
Actual: readmit	54	126
Actual: no readmit	72	648

- (i) (1 %) Compute, to two decimals, the test *sensitivity*, *specificity*, and overall *error rate* of the classifier.
- (ii) (1 %) The care-coordination team argues that “we are missing too many true readmissions.” Indicate the direction (up / down) in which *sensitivity* and *specificity* are expected to move if the classification threshold is lowered from 0.5 to 0.3.
- (iii) (1 %) In the readmission context (where a false negative means a high-risk patient is sent home without targeted follow-up), explain in one sentence which of *sensitivity* and *specificity* you would prioritize, and why.

c) AdaBoost: pseudocode and a small numerical iteration (6 %)

To compare with logistic regression, an AdaBoost classifier with binary weak learners (stumps) is also fit. The classifier returns a label $G(x) \in \{-1, +1\}$ (with +1 encoding readmission).

- (i) (2 %) Write *pseudocode* for the AdaBoost algorithm with M rounds. Make explicit (a) the initialization of the sample weights w_i , (b) the weighted misclassification rate err_m , (c) the classifier weight α_m , (d) the per-observation weight update, and (e) the final ensemble classifier.
- (ii) (2 %) A tiny illustrative example: at iteration $m = 1$ on a training sample of $N = 10$ observations, the initial sample weights are $w_i = 1/10$. The stump $G_1(x)$ misclassifies exactly two of the ten observations. Compute the weighted misclassification rate err_1 , the classifier weight α_1 , and the *updated, un-normalized* weights w_i for (a) a correctly-classified observation and (b) a misclassified one. Round numerics to two decimals.
- (iii) (1 %) On the same test set, the AdaBoost ensemble (with $M = 200$ stumps) achieves sensitivity 0.55, specificity 0.86, and error rate 0.16. Compare these to the logistic regression numbers from (b)(i). Which model would you recommend to the hospital, and on which criterion?
- (iv) (1 %) True or false, with one sentence of justification: “AdaBoost can be derived as a special case of gradient boosting, in which the loss function is the *exponential loss* $L(y, f) = e^{-yf}$.”

d) A neural network classifier and a backprop mini-derivation (8 %)

You also fit a feed-forward neural network with the following architecture (using the standardized version of the six predictors above, plus a derived prev_adm^2 to capture non-linearity, for $p = 7$ inputs):

- input layer of 7 standardized predictors;
- hidden layer of $M = 10$ neurons (with biases, ReLU activations);
- output layer of 1 neuron (with bias, sigmoid activation, returning $\hat{p}$).

- (i) (1 %) How many parameters does this network have in total, *including all bias terms*? Show your layer-by-layer breakdown.
- (ii) (1 %) A neuron in the hidden layer has weights $w = (0.20, -0.50, 1.00, 0.40, -0.60, 0.30, 0.10)$ and bias $b = -0.20$. Its inputs on a given observation are $x = (0.5, -0.5, 2, 1, 0, 1, -1)$. Compute the output of this neuron under a ReLU activation, rounded to two decimals.
- (iii) (3 %) A mini backpropagation derivation. Consider, for a single observation (x_i, y_i) , the simple network

$$z_k = g\left(\alpha_{k0} + \sum_{j=1}^p \alpha_{kj} x_{ij}\right), \quad \hat{f}(x_i) = \beta_0 + \sum_{k=1}^M \beta_k z_k,$$

with squared-error loss $L_i = \frac{1}{2}(y_i - \hat{f}(x_i))^2$ and a generic differentiable activation $g(\cdot)$.

- (a) (1 %) Compute $\partial L_i / \partial \hat{f}(x_i)$.
- (b) (1 %) Using the chain rule and your answer to (a), compute $\partial L_i / \partial \beta_k$ for $k \geq 1$ in terms of z_k , y_i , and $\hat{f}(x_i)$.
- (c) (1 %) Using the chain rule once more, compute $\partial L_i / \partial \alpha_{kj}$ in terms of y_i , $\hat{f}(x_i)$, β_k , x_{ij} , and $g'(v_{ik})$, where $v_{ik} = \alpha_{k0} + \sum_j \alpha_{kj} x_{ij}$ is the pre-activation of hidden unit k .
- (iv) (2 %) Briefly state *two* regularization techniques from this course that you would apply to this network. For each, give a concrete numeric value of any associated hyperparameter where applicable, and one short sentence on what role it plays. (*One of the two must be a technique other than L2 weight decay.*)
- (v) (1 %) A friend trains the same network with mini-batch SGD (batch size 128) and no explicit weight-decay penalty, and observes that it generalizes *better* than the much smaller logistic regression in (a). Briefly state one feature of mini-batch SGD that, according to the lectures, can rationalize this surprising outcome.

End of exam. Total: 10 + 28 + 16 + 22 + 24 = 100 points.